UNIT–I:

Mathematical Logic: Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Tautological Implications, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises, Indirect Method of Proof, Predicate Calculus: Predicates, Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.

UNIT-II:

Set Theory: Sets: Operations on Sets, Principle of Inclusion-Exclusion, Relations: Properties, Operations, Partition and Covering, Transitive Closure, Equivalence, Compatibility and Partial Ordering, Hasse Diagrams, Functions: Bijective, Composition, Inverse, Permutation, and Recursive Functions, Lattice and its Properties.

UNIT-III:

Combinatory and Recurrence Relations: Basis of Counting, Permutations, Permutations with Repetitions, Circular and Restricted Permutations, Combinations, Restricted Combinations, Binomial and Multinomial Coefficients and Theorems.

Recurrence Relations:

Generating Functions, Function of Sequences, Partial Fractions, Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations

UNIT-IV:

Graph Theory:

Basic Concepts, Graph Theory and its Applications, Sub graphs, Graph Representations: Adjacency and Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs,

Unit-V:

Multi Graphs

Multigraphs, Bipartite and Planar Graphs, Euler's Theorem, Graph Coloring and Covering, Chromatic Number, Spanning Trees, Prim's and Kruskal's Algorithms, BFS and DFS Spanning Trees.

Subject: **Discrete Mathematics and Graph Theory**

Name of the Subject coordinator:

UNIT-I (Short Answer Questions. Each 2 Marks)		
Q.No.	Question	
1	Explain about logical connective Biconditional.	
2	Explain about logical equivalence.	
3	Define Duality law.	
4	Define well-formed formula	
5	Define i) tautology ii) contradiction.	
6	Define universal quantifiers	
7	Define existential quantifiers	
8	Define PDNF & PCNF.	
9	Write the minterms of p,q,r	
10	Define Demorgan's laws	

UNIT-I (Long Answer Questions. Each 10 Marks)			
Q.No.	Question	Marks	
1.(i)	Prove that $\{p\rightarrow(q\rightarrow r)\}\rightarrow\{(p\rightarrow q)\rightarrow(p\rightarrow r)\}$is a tautology	5M	
1.(ii)	Prove that for any three propositions p, q, r $\neg [p\vee (q\wedge r)] \leftrightarrow [(p\vee q) \wedge(p\vee r)]$ is a contradiction.	5M	
2.(i)	Construct the truth table for the compound proposition $(p\rightarrow r)\wedge[(q\wedge\neg r)\rightarrow(p\vee r)]$.	5M	
2.(ii)	Define the converse, inverse and contrapositive of the implication and write converse, inverse and contrapositive of the implication "If today is a holiday, then I will go for a movie".	5M	
3.(i)	Prove that for any three proposition p, q, r the conditional $\{(p\rightarrow q) \wedge(q\rightarrow r)\}.\rightarrow(p\rightarrow r)$ is a tautology.	5M	
3.(ii)	Obtain the PCNF of $(\neg p\rightarrow r)\wedge (q\leftrightarrow p)$ and hence find PDNF	5M	
4.(i)	Explain in detail about the logical connectives with examples	5M	
4.(ii)	Prove that $(\neg p\leftrightarrow q) \Leftrightarrow (p\vee q) \wedge \neg (p\wedge q)$.	5M	
5.(i)	Prove that for any three propositions p, q, r $[p\vee (q\vee r)] \wedge\neg q \Rightarrow (p\vee r)$	5M	
5.(ii)	Find the PDNF and PCNF of the formula $p\vee[\neg p\rightarrow\{q\vee(\neg q\rightarrow r)\}]$.	5M	

6.(i)	Show that $r \wedge (p \vee q)$is valid from the premises $p \vee q, q \rightarrow r, p \rightarrow m, \neg m$	5M	
6.(ii)	Verify the validity of the following argument. All men are mortal. Socrates is a man. Therefore, Socrates is mortal.	5M	
7.(i)	Show that $S \vee R$ is tautologically implied by P∨Q , P →R , Q→S.	5M	
7.(ii)	Establish the validity of the following arguments. All integers are rational number. Some integers are power of 2. Therefore, Some Rational numbers are Powers of 2.	5M	
8.(i)	Show that the premises $a \rightarrow (b \rightarrow c)$, $d \rightarrow (b \wedge \neg c)$, $a \wedge d$ are inconsistent.	5M	
8.(ii)	Test the validity of the following arguments Sonia is watching TV . If Sonia is watching TV, then she is not studying. If she is not studying, then her father will not buy her scooty. Therefore, Sonia father will not buy a scooty	5M	
9.(i)	If there was a meeting, then catching the bus was difficult. If they arrived on time then catching the bus was not difficult. They arrived on time. Therefore there was no meeting. Show that the statements constitute a valid argument5	5M	
9.(ii)	Symbolize the following argument and check for their validity: "Every living thing is a Plant or an animal. David's dog is a live and it is not plant. All animals have hearts. Hence David's dog has a heart.	5M	
10.(i)	Show that the following premises are inconsistent If Jack misses many classes because of illness, then he fails high school. If Jack fails high school, then he is uneducated. If Jack reads lot of books, then he is not uneducated. Jack misses many classes because of illness and reads a lot of books.	5M	
10.(ii)	Show that $(x)[P(x) \rightarrow Q(x)] \wedge (\exists x)P(x) \Rightarrow (\exists x)Q(x)$.5	5M	

Q.No.	Question
1	Write the formula for Principle of Inclusion and Exclusion for three sets A, B, C
2	Given A = {1, 2} and B = $\{a, b, c\}$ then find i) A × B ii)B × B.
3	Let A = {a, b, c, d} and R is a relation on A that has the matrix $M_R = [1\,0\,1\,0\,0\,1\,0\,0\,1\,1\,1\,0\,0\,1\,0\,1]$ write relation R for the matrix M_R.
4	Let A= {a, b, c} , B = {1, 2, 3} and the relations R = {(a,1), (b,1), (c,2), (c,3)}, S = {(a,1), (a,2), (b,1),(b,2)} from A to B then determine $\underline{R}$, $\underline{S}$.
5	Define equivalence relation.
6	Define Bijective function.
7	If f = $(1\,2\,3\,4\,5\,2\,4\,3\,1\,5)$then show that $fof^{-1} = f^{-1}of$= I.
8	Define cyclic permutation.
9	Define Recursive formula.
10	Define distributive Lattice.

Q.No.	Question	Marks
1.(i)	If A = {1, 2, 3, 4, 5}, B = {2, 4, 6, 8}, C = {1, 2, 6},find $(i)A - B$ (ii) $B - A$ $iii)(A \cup B) - (A \cap C)$	5M
1. (ii)	Let the permutations of the elements {1, 2, 3, 4, 5} be given by $\alpha = (1\,2\,3\,4\,5\,2\,3\,1\,4\,5)$ $\beta = (1\,2\,3\,4\,5\,1\,2\,3\,5\,4)$ $\gamma = (1\,2\,3\,4\,5\,5\,4\,3\,1\,2)$ $\delta = (1\,2\,3\,4\,5\,3\,2\,1\,5\,4)$ Find $\alpha\beta, \beta\alpha, \alpha^2, \gamma\beta, \beta\delta$.	5M
2.(i)	A sample of 100 logic chips , 23 have defect D1, 26 have a defect on D2, 30 have defect D3 , 7 have defect on D1 and D2, 8 have defect on D1 and D3, 10 have defect on D2 and D3 and 3 have all the three defects. Find the number of chips having i) At least one defect, ii) No defect?	5M
2.(ii)	Let $f = (1\,2\,3\,4\,2\,4\,1\,3)$ and $g = (1\,2\,3\,4\,4\,1\,2\,3)$ then find ff, fg, gf and gg.	5M
3.(i)	Let$A$ = {1, 2, 3} and B = {1,2,3,4}.The relations R and S from A to B are represented by the following matrices. Determine the relations $\underline{R}$, $R \cup S$, $R \cap S$and $R - S$, $\underline{S}$. $M_R = [1\,0\,1\,0\,0\,0\,0\,1\,1\,1\,1\,0]$ and $M_S = [1\,1\,1\,1\,0\,0\,0\,1\,0\,1\,0\,1]$	5M
3.(ii)	Let f(x) = x + 2, g(x) = x - 2, h(x) = 3x $\forall$ x ϵ R where R is the set of real numbers for gof, fog, gog, fof, foh, hog, hof and fogoh.	5M
4.(i)	How many positive integers not exceeding 2000 are divisible by 7 or 11	5M
4.(ii)	Let f:$R \to R$, g:$R \to R$ where R is the set of real numbers be given by f(x)=x + 2, g(x)=x^2. Find fof, fog, gof, gog.	5M
5.(i)	5. Let$A$ = {1, 2, 3, 4, 6, 8, 12} On A, define the partial ordering relation R by aRb if and only if $a|b$. (i)Draw the Hasse diagram for R ii) Write down the relation matrix for R.	5M
5.(ii)	Let A = B = {a, b, c, d} and R = {(a,a), (a,c), (b,c), (c,a), (d,b), (d,d)} and S = {(a,b), (b,c), (c,a), (c,b), (d,c)} compute M_R and M_S also Draw the digraphs	5M

6.(i)	Define power set. Find the power set of A, where $A = \{1, 2, \{1, 2\}\}$	5M
6.(ii)	Find the domain and co domain of $f(x) = \frac{2x+3}{4x-5}$ so that it is Bijective. Hence find the inverse of f(x).	5M
7.(i)	Let A = {1, 2} and B = {3, 4}. Write A × B. How many relations are there from A to B. List them	5M
7.(ii)	Consider the following recursive function definition: If y < x, then f(x, y) = 0; If x ≤ y, then f(x, y) = f(x, y - x) + 2. Find the value of f(5,487)	5M
8.(i)	Let A = {1, 2, 3, 4, 6} and R be a relation on A defined by aRb if and only if a is a multiple of b. Represent the relation R as a matrix and draw its digraph.	5M
8.(ii)	Define Lattice. Also write the properties of lattices.	5M
9.(i)	Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(x, y) / x - y \; is divisible by \; 3\}$. Show that R is an equivalence relation. Draw the graph of R.	5M
9.(ii)	Let A = {1, 2, 3, 4} and let S = {(1,2), (2,3), (3,4)} be a relation on A. Find the transitive closure of S.	5M
10.(i)	Define compatible relation. Explain maximal compatible blocks with an example.	5M
10.(ii)	Write the different types of functions Explain with example.	5M

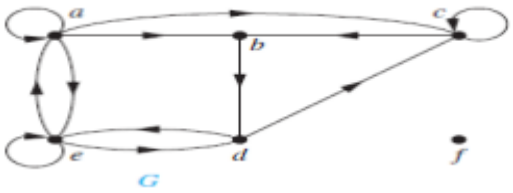
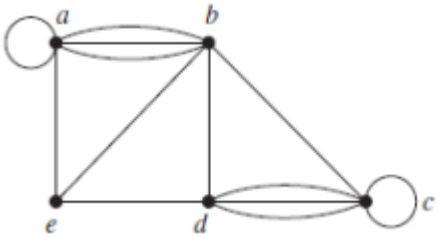
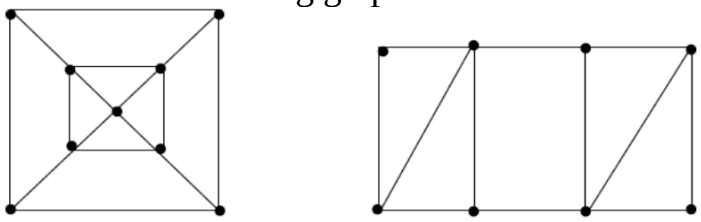
Q.No.	Question
1	Find the number of permutations of the letters of the word SUCCESS.
2	In how many different ways can a committee of 5 teachers and 4 students be selected from 9 teachers and 15 students?
3	Find the coefficient of x^9y^3 in the expansion of $(x + 2y)^{12}$.
4	Evaluate(12 5 3 2 2).
5	Find the coefficient of xyz^5 in the expansion of $(x + y + z)^7$.

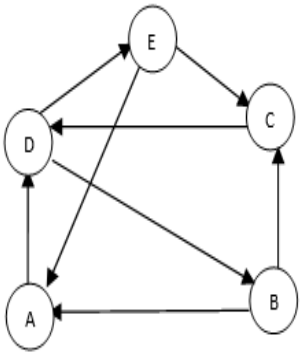
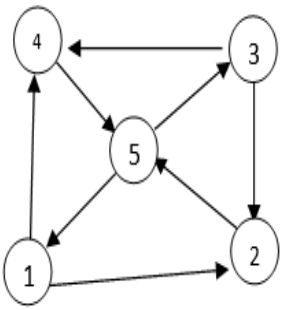
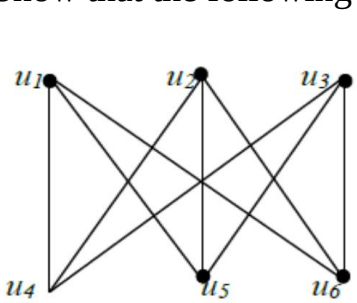
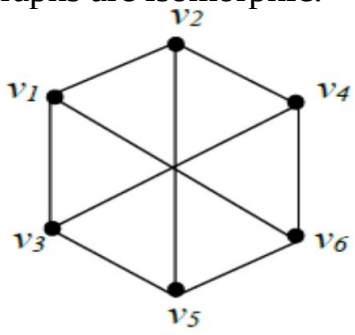
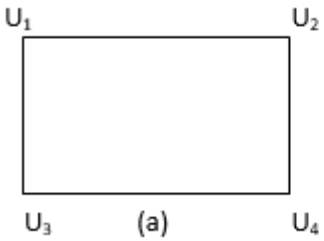
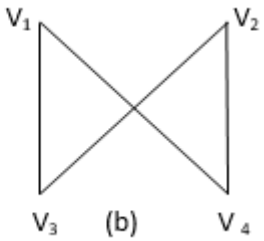
6	In how many ways can we distribute 10 identical marbles among 6 distinct containers?
7	Find the generating function for the sequence 1,-2, 3,-4, -----.
8	The two roots of k second order linear homogeneous Recurrence relations are real and distinct, say k_1 and k_2 then general solution is.
9	Write Fibonacci relation and sequence.
10	Write Linear Non Homogeneous Recurrence relation of second and higher order and also write its general solution.

Q.No.	Question	Marks
1.(i)	Find the number of Permutations of the letters of the word **MASSASAUGA**. In how many ways of these all A's are together? How many of them begin with S?	5M
1.(ii)	How many committees of 5 members with a given chair person can be selected from the 12 persons.	5M
2.(i)	Find the coefficient of $x^{101}y^{99}$ in the expansion of $(2x-3y)^{200}$	5M
2.(ii)	Find the number of integer solutions of $x_1+x_2+x_3+x_4+x_5 = 30\ where\ x_1 \geq 2, x_2 \geq 3, x_3 \geq 4, x_4 \geq 2, x_5 \geq 0$	5M
3.(i)	Determine the coefficient of $x^3y^3z^2$ in the expansion of $(2x-3y+5z)^8$	5M
3.(ii)	Find the number of non-negative integer Solutions $x_1+x_2+x_3+x_4+x_5+x_6 < 10$	5M
4.	A woman has 11 close relatives and she wishes to invite 5 of them to dinner. In how many ways can she invite them in the following situations: (i) There is no restriction on the choice. (ii) Two particular persons will not attend separately. (iii) Two particular persons will not attend together.	10M
5.(i)	How many Positive integers n can we form using the digits 3, 4, 4, 5, 5, 6, 7. if we want $'n'$ to Exceed 5,000,000	5M
5.(ii)	A certain question paper contains three parts A, B, C with four questions in part A, five questions in part B and six questions in part C. It is required to answer seven questions selecting at least two questions from each part. In how many ways can a student select his seven questions from answering?	5M
6.(i)	Find the generating functions for the following sequence $i)\ 1,2,3,4,\ldots\ldots$ $ii)\ 1,\frac{1}{1!},\frac{1}{2!},\frac{1}{3!},\ldots\ldots$	5M
6.(ii)	Solve the R.R $F_{n+2} = F_{n+1} + F_n\ for\ n \geq 0, given\ F_0 = 0, F_1 = 1$	5M
7.(i)	Find the generating functions for the following sequence $i)\ 0,1,-2,3,-4,\ldots\ldots$ $ii)\ 0,2,6,12,20,30,42\ldots$	5M
7.(ii)	Solve the R.R $a_n - 7a_{n-1} + 10a_{n-2} = 4^n, n \geq 2, a_0 = 8, a_1 = 36.$	5M
8.(i)	Find the coefficient of x^{27} in the function $(x^4+x^5+x^6+\cdots.)^5.$	5M
8.(ii)	Using method of characteristic roots solve $a_n - 7a_{n-1} + 16a_{n-2} - 12a_{n-3} = 0, n \geq 3$ With the initial conditions $a_0 = 1, a_1 = 4, a_2 = 8$	5M
9.	Solve the R.R $a_n + 4a_{n-1} + 4a_{n-2} = 8\ for\ n \geq 2\ and\ a_0 = 1, a_1 = 2.$	10M
10.	Solve the R.R $a_{n+2} - 10a_{n+1} + 21a_n = 3n^2 - 2\ for\ n \geq 0.$	10M

UNIT-IV (Short Answer Questions. Each 2 Marks)	
Q.No.	Question
1	Define order and size of a graph.
2	Define isolated and pendent vertex.
3	State Handshaking property.
4	Define complement of a graph.
5	Define walk.
6	Define trail.
7	Define connected and disconnected graphs.

8	Define Euler circuit and Euler Trail.
9	Draw the diagram of 3-D Hyper cube.
10	Define peterson graph and Draw its diagram.

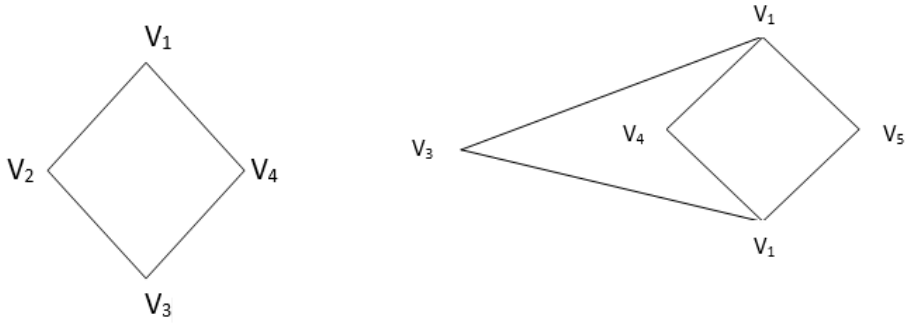
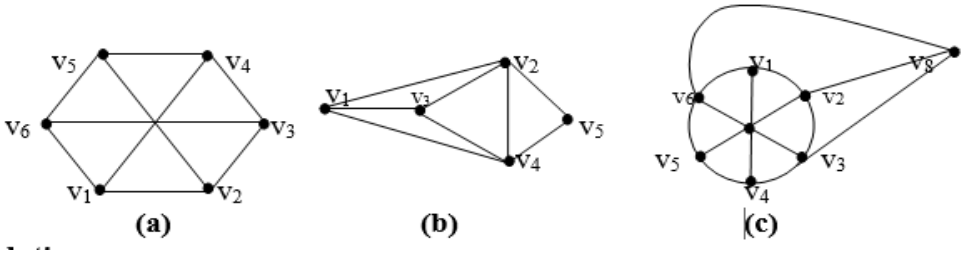
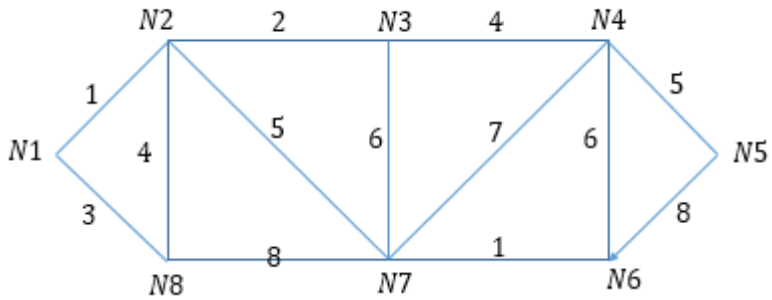
Q.No.	Question	Marks
1.(i)	Define the terms Path, Simple Path, Circuit, Cycle and Simple Graph in the graphs and give an examples.	5M
1. (ii)	Find the in-degree and out-degree of each vertex in the graph G with directed edges  G	5M
2.(i)	Explain adjacency matrix and incidence matrix for a graph with suitable example.	5M
2.(ii)	If G is a simple graph in which every vertex has degree at least k, prove that G contains a path of length at least k	5M
3.(i)	Explain Euler Path, Euler Circuit and Euler graph in a multigraph with suitable examples.	5M
3.(ii)	Explain Hamilton Cycle, Hamilton Path and Hamilton graph with suitable examples.	5M
4.(i)	Find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph. Identify all isolated and pendant vertices. 	5M
4.(ii)	Prove that the following graphs are Hamiltonian but not Eulerian  (a) (b)	5M
5.(i)	Write about the basic rules for constructing Hamiltonian Cycles.	5M
5.(ii)	Give an example of a graph which is Eulerian but not Hamiltonian.	5M
6.	Show that the following two graphs are isomorphic.	10M

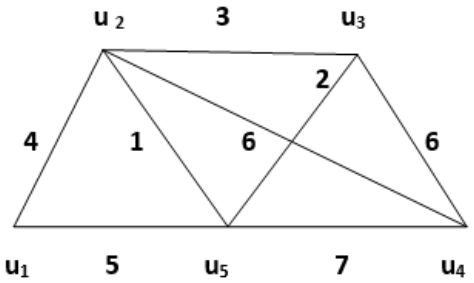
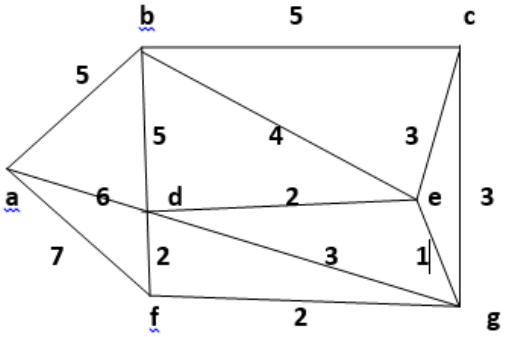
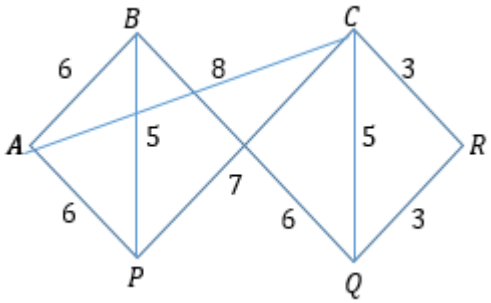
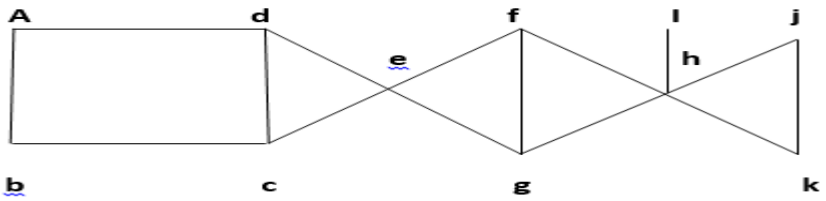
	 	
7	Show that the following two graphs are isomorphic.  *(a)*  *(b)*	10M
8.	Show that the following two graphs are isomorphic  	10M
9.	Define Sub graph, spanning sub graph, induced sub graph, disjoint and vertex disjoint.	10M
10.	Define isomorphism of two graphs. Explain with examples.	10M

UNIT-V (Short Answer Questions. Each 2 Marks)		
Q.No.	Question	
1	Define chromatic number?	
2	Define graph coloring.	
3	State Euler's formula.	
4	Draw K_1,K_2 and K_4 planar graphs.	
5	Define Five color theorem.	
6	Define Spanning tree.	
7	Define minimal spanning tree with an example.	

8	Define Trivial and non-trivial tree.	
9	Define complete Binary tree.	
10	Define Binary tree.	

UNIT-V (Long Answer Questions. Each 10 Marks)

Q.No.	Question	Marks	
1.(i)	Prove that c_6 is a bipartite graph	5M	
1. (ii)	How many vertices and how many edges are there in Complete Bipartite graphs $k_{4,7}$ and $k_{7,11}$	5M	
2.	Explain i)planar graph ii) Regular graph iii) Bipartite graph iv) Complete Bipartite graph v) Multi graph with examples	10M	
3.	Show that the Bipartite graphs $K_{2,2}$ and $K_{2,3}$are planar graphs. 	10M	
4.	Show that the complete graph k_5 and complete bipartite graph $k_{3,3}$ are not planar.	10M	
5.	Find the chromatic number of each of the following graphs. 	10M	
6.	Find minimal spanning tree for the following graph using Prim's algorithm 	10M	

7.	Find the minimal spanning tree for the following graph using Prim's algorithm 	10M	
8.(i)	Using Krushkal's algorithm find a minimal spanning tree for the following graph. 	10M	
9.(i)	Using Krushkal's algorithm find a minimal spanning tree for the following graph 	10M	
10.(i)	Using BFS & DFS algorithm, find the spanning tree for the following graph. 	10M	

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